Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics

Constant Velocity Lab

Criterion C — Processing and evaluating • formative

Newton's idea, tested

In the Newton's Laws lesson, the skydiver showed you the key idea of Newton's First and Second Laws: **when the forces on something are balanced, its speed stays constant.** Today you test that idea on a robot.

Once the robot is moving, the push from its wheels balances the friction against it. So Newton's laws make a prediction you can check with a stopwatch:

> **Prediction:** the forces on the moving robot are **balanced**, so it keeps a **constant speed**. It travels **equal distances in equal times**, and its distance–time graph is a **straight line**.

Constant velocity means a steady speed in a straight line — the "balanced" row of the Newton's First Law table. You will measure that speed with the formula from the speed lesson: speed = distance ÷ time.

The method

One robot for the whole class

Kit: the robot; a 3 m track taped on the floor, with a line every 0.5 m from the start line (0 m) to 3.0 m; six stopwatches.

Jobs for each run

- **Driver** — switches the robot on and sets it down 30 cm behind the start line, pointing straight along the track. The run-up means it is already at its steady speed when timing begins.
- **Starter** — stands at the start line and calls **"Go!"** the instant the front of the robot crosses it.
- **Six timers** — one at each line from 0.5 m to 3.0 m. Start your stopwatch on "Go!" and stop it the instant the front of the robot crosses *your* line.
- **Scribe** — writes each timer's time on the board, to 1 decimal place.

Steps

1. Run the robot. Each timer reads out their time and the scribe writes it in the **Run 1** column on the board.
2. Swap jobs and run it again (**Run 2**), then once more (**Run 3**), so everyone times at least once.
3. Copy all eighteen times from the board into your own table.

Check: each time on the board is bigger than the time at the line before it — the robot reaches the far lines later.

Care: stand beside the track, outside the tape, so the robot's path stays clear.

C1 Your results

The **0 m row is filled in for you**: every stopwatch starts as the robot crosses the start line. Work out each **average** — add the three times and divide by 3 — and round it to 1 decimal place.

Example data only — a robot covering about 0.5 m every 2 seconds, with the scatter hand timing gives. Your class's times depend on the robot; the pattern is what to compare.

Distance / m	Run 1 time / s	Run 2 time / s	Run 3 time / s	Average time / s
0	0	0	0	**0**
0.5	2.1	1.9	2.2	**2.1**
1.0	4.0	4.2	3.9	**4.0**
1.5	6.1	5.9	6.0	**6.0**
2.0	8.2	7.9	8.0	**8.0**
2.5	9.9	10.2	10.1	**10.1**
3.0	12.1	11.9	12.2	**12.1**

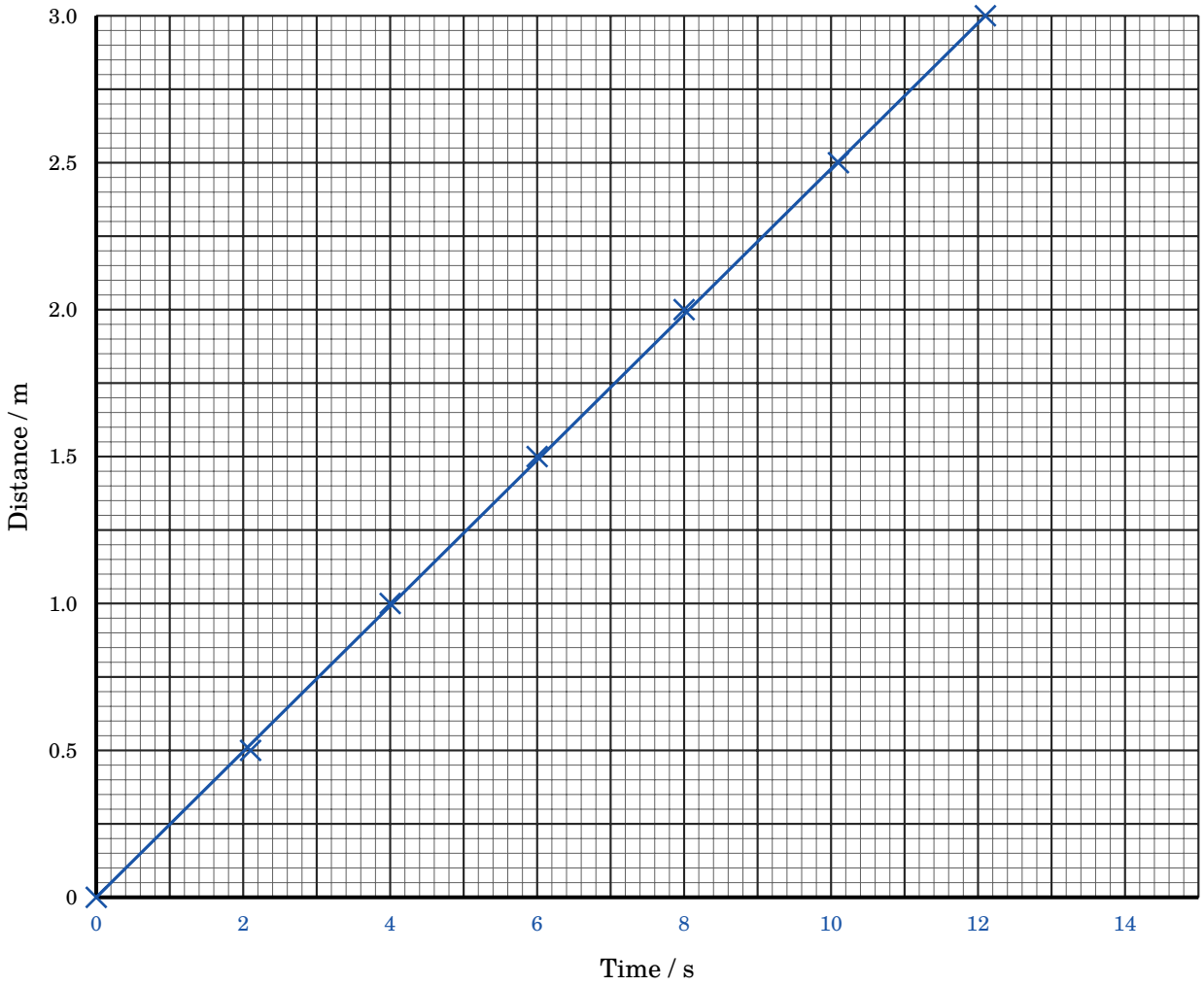
Check: each average lies between the smallest and the largest of its three times.

C1 Your graph

A **distance–time graph**: distance up the side, time along the bottom. The distance axis is numbered for you.

1. **Number the time axis.** Choose **0.5 s, 1 s, 2 s or 5 s** for each big square — the smallest that lets your longest average time fit on the axis.
2. **Plot** each average time against its distance as a small, neat cross. The first point is $(0,\ 0)$.
3. If your points lie close to a straight line, draw **one straight line of best fit with a ruler**. If they curve, draw **one smooth curve**.

Check: your line passes close to every point, with about as many points above it as below.



With the example data, 1 s per big square fits: the longest average is 12.1 s on a 15-square axis.

C2 What your graph shows

Circle the words that match your graph, and fill in the blanks.

Choose the words that fit your graph

My graph is a **straight line** / **curve**, so the robot travelled **equal** distances in **equal** times.

This **supports** / **does not support** the prediction that the forces on the moving robot are **balanced**.

Back it up with **two numbers from your own table**.

Model answer: The robot reached 1.0 m at 4.0 s and 2.0 m at 8.0 s — twice the distance in twice the time.

Now work out the robot's **speed** over the whole track, using your average time at 3.0 m. Give it to 2 decimal places.

speed = distance ÷ time = 3.0 m ÷ **12.1** s = **0.25** m/s

Check: speed × time gives you back 3.0 m.

C4–C5 How good were your times?

Write it like this

The hardest thing to time exactly was _____.

To make the times more accurate, I would _____.

Model answer: The hardest thing to time exactly was the instant the robot crossed each line, because every timer had to react twice — once to "Go!" and once to the robot. To make the times more accurate, I would film each run with a phone beside the track and read the times off the video frame by frame.

To go further — equal distances in equal times?

Use your **average** times. For each 0.5 m stretch of track, subtract the time at the start of the stretch from the time at the end of it. Then work out the robot's speed on each stretch: 0.5 m ÷ time, to 2 decimal places.

Stretch / m	0–0.5	0.5–1.0	1.0–1.5	1.5–2.0	2.0–2.5	2.5–3.0
Time / s	**2.1**	**1.9**	**2.0**	**2.0**	**2.1**	**2.0**
Speed / m/s	**0.24**	**0.26**	**0.25**	**0.25**	**0.24**	**0.25**

What do you notice about the six speeds, and how do they compare with your speed for the whole track?

Model answer: All six speeds are about 0.25 m/s, the same as the speed over the whole 3.0 m. The robot covered equal distances in equal times, so its speed stayed constant all the way along the track.